

Subgroup Performance

Project: Responsible Plant Classifier

1. Subgroup Performance

Teach the learning cycle

Primary target: Calculate subgroup results with sample sizes

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Look for: Compare training and validation evidence for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

Learning Curves

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Look for: Read training-size learning curves

Reference example, not the only acceptable wording: Teacher explanation: $\text{Mean}(5+7+6)/3=6$; range $7-5=2$.

Metrics Match Consequences

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a declared score threshold including equality

Reference example, not the only acceptable wording: Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the subgroup gap. What causal claim is not supported?

Concept to explain

Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

Term · Meaning

Term: subgroup

Meaning: meaningful slice of evaluation cases

Term: performance gap

Meaning: difference between subgroup metrics

Term: denominator

Meaning: number of cases behind a rate

Term: intersection

Meaning: cases belonging to multiple slices

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on subgroup evaluation. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Different error patterns in different evaluated conditions or groups.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain subgroup performance using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Calculate subgroup accuracy and recall gaps with adequate denominators.

MODEL IDEA Overall metrics can conceal uneven behavior across meaningful conditions or user groups.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Daylight: 18/20 correct = 90%; evening: 10/20 = 50%. Accuracy gap = 90% – 50% = 40 percentage points.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Daylight: 18/20 correct = 90%; evening: 10/20 = 50%.

STEP 2 Accuracy gap = 90% – 50% = 40 percentage points.

STEP 3 Evening spotted recall is 4/10 = 40%, versus daylight 8/10 = 80%.

STEP 4 Inspect representation and image quality; report counts and uncertainty before repairing.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

A has 18/20 correct; B has 6/10.

Compute subgroup results and evidence limits.

Exact explanation to model

90% versus 60%, gap 30 percentage points. Report denominators and investigate collection/label differences without assuming an inherent group cause.

Explain the added worked example

A single overall score can hide different experiences. Define relevant groups before evaluation and report counts as well as rates. This classroom exercise uses fictional sensor conditions, not personal identities. Real evaluations involving people require appropriate permissions, privacy and contextual expertise.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines subgroup evaluation.

Sensor group A has 2 errors in 20 examples and group B has 3 in 5. Error rates are 10 and 60 percent; pooled error is 5/25=20 percent. The group difference is 50 percentage points. B has only five examples, so its rate is sensitive to each individual result.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Key terms

Term · Meaning

Term: Subgroup

Meaning: A defined subset of the evaluated examples

Term: Disaggregation

Meaning: Reporting results separately for defined groups

Term: Coverage

Meaning: Which groups and conditions are represented

Term: Disparity

Meaning: An observed difference between group results

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Subgroup evaluation at a glance

Part · Evidence or action · Interpretation

Part: Group A

Evidence or action: 2 errors of 20

Interpretation: 10 percent

Part: Group B

Evidence or action: 3 errors of 5

Interpretation: 60 percent

Part: Pooled

Evidence or action: 5 errors of 25

Interpretation: 20 percent

Changing one B error to a correct result changes its rate from 60 to 40 percent. Small samples can be unstable without making the observed errors unimportant. A missing group has no measured rate, not zero error. Intersections of conditions may be even smaller.

How does correcting one B error affect its rate? Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Subgroup Performance

Calculate subgroup and pooled error rates without letting the larger group hide a gap.

The facts we will use

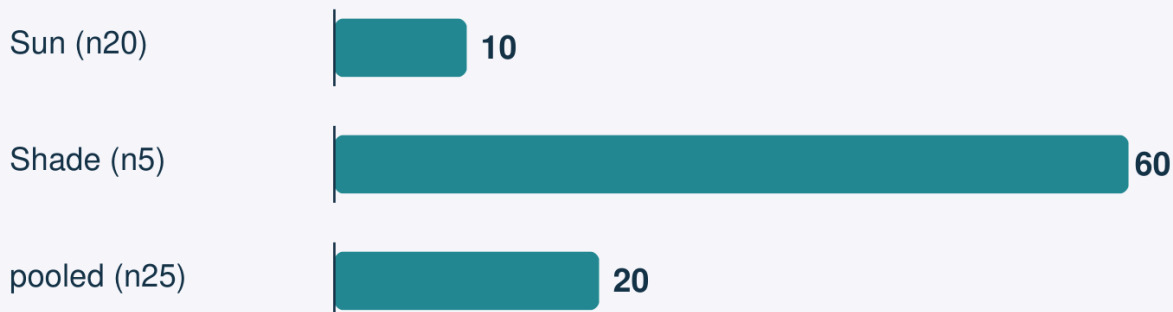
Group Sun: 2 errors in 20 photos; Shade: 3 errors in 5. Same error definition; related plant views are documented.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/31

Subgroup error rates

percent



Bar length shows the supplied value; the marked baseline is zero.

Subgroup Performance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Sun rate

Sun error= $2/20=10\%$.



2. Shade rate

Shade error= $3/5=60\%$.



3. Pooled rate

Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference.

CHANGE ONE THING

Shade gains 5 new cases with no errors.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Sun rate

What happens and why: Sun error= $2/20=10\%$.

Step: 2. Shade rate

What happens and why: Shade error= $3/5=60\%$.

Step: 3. Pooled rate

What happens and why: Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Total error= $5/25=20\%$, which hides the 50-percentage-point subgroup difference. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Small subgroup samples have high variability. An observed gap does not by itself establish discrimination or a causal explanation.

Change one thing in the pictured case

Change: Shade gains 5 new cases with no errors.

Prediction: Shade error becomes $3/10=30\%$; pooled error becomes $5/30\approx 16.7\%$.

Reason: Extra evaluated cases change denominators; existing errors have not disappeared.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A has $18/20$ correct; B has $6/10$.

Write the question to explain: Compute subgroup results and evidence limits.

Reveal after predictions: 90% versus 60%, gap 30 percentage points. Report denominators and investigate collection/label differences without assuming an inherent group cause.

Keep for an independent attempt: Accuracy is $18/20$ in bright light and $6/10$ in dim light. Compute both rates and the percentage-point gap.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: B becomes $2/5=40$ percent, a 20-percentage-point change from one case. Exit response: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: What can a pooled error rate hide? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify subgroup in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: calculate group rates; check sensitivity. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between subgroup and disaggregation. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Group C has no evaluated cases. What is its measured error rate?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Calculate subgroup results with sample sizes; Distinguish missing evidence from equal performance. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: How does correcting one B error affect its rate?

Teacher check: Teacher explanation: B becomes $2/5=40$ percent, a 20-percentage-point change from one case.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Subgroup evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: A $2/20=10$ percent; B $3/5=60$ percent; pooled $5/25=20$ percent, a 50-point group gap.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why should the report retain subgroup rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

The 20% original total is dominated by Sun's larger sample and understates the observed Shade difficulty. Include counts, conditions and uncertainty. Changed-case check: Shade error becomes $3/10=30\%$; pooled error becomes $5/30\approx 16.7\%$. Extra evaluated cases change denominators; existing errors have not disappeared.

Changed-case reasoning: Shade error becomes $3/10=30\%$; pooled error becomes $5/30\approx 16.7\%$. Extra evaluated cases change denominators; existing errors have not disappeared.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Slice Audit: compute metrics and denominators for lighting, species, and camera; flag small samples and prioritize one gap.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A good overall result guarantees acceptable performance for every subgroup.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: A $2/20=10$ percent; B $3/5=60$ percent; pooled $5/25=20$ percent, a 50-point group gap.

A stronger claim makes a stronger project: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: X 10 percent; Y 50 percent; pooled $3/14$ about 21.4 percent. Y has only four examples, and the table does not establish a cause.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner

restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

90% versus 60%, gap 30 percentage points. Report denominators and investigate collection/label differences without assuming an inherent group cause.

Independent question

Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.

Expected independent answer

Bright: 90%; dim: 60%; gap: 30 percentage points. Report denominators and uncertainty from small samples before making broad claims.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “The difference is 30 percent of all examples rather than 30 percentage points between rates.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Overall= $24/30=80\%$; A= 90% , B= 60% , gap=30 percentage points. These data do not prove the device itself causes the difference; lighting, leaf mix or labels may differ. Investigate with comparable cases and preserve small-sample uncertainty.

Explanations for the additional workbook examples

W1 Calculate group rates

Calculate the worked group and pooled rates.

Teacher explanation: A $2/20=10$ percent; B $3/5=60$ percent; pooled $5/25=20$ percent, a 50-point group gap.

W2 Check sensitivity

How does correcting one B error affect its rate?

Teacher explanation: B becomes $2/5=40$ percent, a 20-percentage-point change from one case.

W3 Report missing coverage

Group C has no evaluated cases. What is its measured error rate?

Teacher explanation: Unavailable because the denominator is zero; report missing coverage and do not substitute zero.

W4 Apply independently

Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

Teacher explanation: X 10 percent; Y 50 percent; pooled $3/14$ about 21.4 percent. Y has only four examples, and the table does not establish a cause.

W5 Avoid a fairness shortcut

Why does equal accuracy not settle whether use is fair?

Teacher explanation: Error types, consequences, access, labels and context can differ; a single equality does not establish acceptable treatment.

Assess independent transfer

Selected case: Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

Selected case answer

Teacher explanation: X 10 percent; Y 50 percent; pooled $3/14$ about 21.4 percent. Y has only four examples, and the table does not establish a cause.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Add subgroup tables and a minimum acceptable threshold to the Plant Classifier release criteria.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 32: Proxy Features and Fairness. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use subgroup and disaggregation in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: X 10 percent; Y 50 percent; pooled 3/14 about 21.4 percent. Y has only four examples, and the table does not establish a cause. Fresh retry X 2/8 and Y 1/2:25 and 50 percent, pooled 3/10=30 percent. Require denominators, the small Y sample and no causal inference. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Proxy features and fairness. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/32, ai-4/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.